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Topical Review

# Local Order Parameters For Ice Polymorphs: A Unified Comparison And Practical Guide

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**Abstract**

Molecular simulations of water and ice depend on local order parameters to distinguish solid from liquid, one ice polymorph from another, and to track nucleation, growth and interfaces at single-molecule resolution. Dozens of such parameters have been proposed, from tetrahedral order and the local structure index to Steinhardt bond-orientational invariants, CHILL+, template matching and machine-learned descriptors, yet they are usually compared using values taken from different codes, cutoffs and water models. Here we review these parameters within a single taxonomy organized by the structural information they encode, and benchmark [TODO: N] of them with one consistent, open implementation (SolOrder) on [TODO: N] crystalline phases, amorphous ices and liquid water, in both ideal and thermalized configurations. We quantify pairwise phase separability, sensitivity to neighbor definitions, computational cost and suitability as collective variables, and discuss where data-driven descriptors and learned classifiers improve on physically motivated parameters. We conclude with practical recommendations for choosing an order parameter for a given question.

# Table of Contents

**1 Introduction** **1**
1.1 Scope and organization . . . . . 1

**2 The ice landscape in brief** **2**
2.1 Phase diagram and stability . . . . . 2
2.2 Structural signatures relevant to local order . . . . . 2
2.3 Amorphous ices, clathrates and liquid water . . . . . 2

**3 A taxonomy of local order parameters** **3**
3.1 Common conventions . . . . . 3
3.2 Radial and density-based parameters . . . . . 3
3.3 Angular and tetrahedral parameters . . . . . 3
3.4 Hydrogen-bond and topology-based parameters . . . . . 4
3.5 Bond-orientational order . . . . . 4
3.6 Template and symmetry-based parameters . . . . . 4
3.7 Molecular and electrostatic parameters . . . . . 4
3.8 Summary table . . . . . 4

**4 Descriptors and machine-learned approaches** **6**
4.1 High-dimensional descriptors . . . . . 6
4.2 Supervised classifiers . . . . . 6
4.3 Unsupervised and dimensionality-reduction approaches . . . . . 6
4.4 Data-driven collective variables . . . . . 7

**5 A unified benchmark** **8**
5.1 Datasets . . . . . 8
5.2 Separability metric . . . . . 9
5.3 Phase-by-phase separability . . . . . 9
5.4 Ice labelers against ground truth . . . . . 12
5.5 Sensitivity to neighbor definition and water model . . . . . 13

5.6 Computational cost . . . . . 13
5.7 Application: following an interface or nucleation event . . . . . 14

**6 Practical guide** **15**
6.1 Software . . . . . 15
6.2 Reproducibility checklist . . . . . 16

**7 Outlook** **17**

# List of Figures

5.1 Number of parameters (of 39) that separate each pair of ideal crystals with a balanced accuracy of at least 0.95. The full parameter × pair matrix is in the Supplementary Information (S1_separability_ideal_all_pairs). 10

5.2 Best single-threshold balanced accuracy for ice versus liquid, Ih versus Ic, and hexagonal versus cubic layers in stacking-disordered ice, TIP4P/Ice at 220 K. Left: thermal frames; right: the same frames after energy minimization. . . . . 11

5.3 Hierarchical clustering of parameters by $1 - |\rho|$, with $\rho$ the Spearman correlation over thermal Ih, Ic and liquid molecules at 220 K. The dashed line marks $|\rho| = 0.8$. . . . . 12

5.4 Fraction of interior seed molecules labeled with the correct polytype by CHILL, CHILL+ and PTM, for Ih, Ic and stacking-disordered (Isd) seeds of $N$ molecules in TIP4P/Ice liquid at 220 K. Error bars: 95% bootstrap intervals over frames. . . . . 13

# List of Tables

3.1 Local order parameters in the benchmark, with the settings used there. All O–O first-shell cutoffs are 3.5 Å. [TODO: add cost column after the scaling runs.] ?: reference to add. . . . . 5

6.1 Provisional recommendations from the benchmark so far (ideal crystals and TIP4P/Ice at 220 K). [TODO: revise with the thermalized high-pressure phases.] . . . . . 15

# Chapter 1

# Introduction

*Plan: Why local structure matters in water and ice simulation: nucleation, polymorph selection, interfaces, hydrates. Define a local order parameter (LOP) and separate the three jobs it can do: classification (labeling molecules), description (distributions as fingerprints), and collective variables (driving enhanced sampling). State the gap: existing comparisons mix codes, cutoffs and water models. State the contributions: taxonomy, unified benchmark, practical guide, open software.*
*Reuse: intro/introduction.tex paragraphs 4–5 (phase-space complexity, Hansen2021 phase diagram figure); drop the per-phase descriptions in paragraphs 1–2, which move to Chapter 2.*

[TODO: literature update 2023–2026: ML potentials, data-driven CVs for ice nucleation, recent order-parameter reviews to position against.]

## 1.1 Scope and organization

*Plan: What is covered (bulk crystalline ices, clathrate frameworks, amorphous ices, liquid; local per-molecule parameters) and what is not (global crystallographic analysis, spectroscopic observables, electronic structure).*

# Chapter 2

# The ice landscape in brief

*Plan: Short narrative of the phase diagram plus one master table; no per-phase subsections.*
*Reuse: icephases/icephases.tex; condense into the table and keep 3–4 representative structure figures (Ih, Ic, VI, sI).*

## 2.1 Phase diagram and stability

*Plan: Experimental vs. model (TIP4P/Ice, TIP4P/2005, mW) phase diagrams; proton order–disorder pairs; metastable phases; current count of crystalline phases.*
*Reuse: intro/phase.png, conclude/tip4p.png.*

[TODO: verify phase count and add Ice XIX and later reports.]

## 2.2 Structural signatures relevant to local order

*Plan: Master table, one row per phase: space group, molecules per cell, density, proton order, first-shell coordination, second-shell interpenetration, ring statistics, stability window, water-model caveats. Argue which features a LOP can and cannot see (e.g. proton-order partners share oxygen sublattices).*
*SolOrder: docs/validation/genice_summary.csv for density and CN columns.*

## 2.3 Amorphous ices, clathrates and liquid water

*Plan: LDA/HDA/VHDA, sI/sII/sH frameworks, liquid structure and the two-state picture; why these are the hardest cases for LOPs.*
*Reuse: icephases/icephases.tex section "Clathrates, LDA and HDA".*

# Chapter 3

# A taxonomy of local order parameters

*Plan: Every parameter gets: definition, neighbor convention, range and limits, the phases it was designed for, known pitfalls. Keep equations; cut narrative. End with a summary table (Table 3.1).*
*Reuse: background/background.tex, reorganized into the groups below.*

## 3.1 Common conventions

*Plan: Neighbor definitions (cutoff, N nearest, Voronoi, H-bond graph); oxygen-only vs. molecular parameters; local averaging (Lechner–Dellago style); periodic boundaries; smooth vs. discontinuous forms and why it matters for collective variables.*
*SolOrder: neighbors.py; averaged=True options.*

## 3.2 Radial and density-based parameters

*Plan: Density, cutoff and smooth coordination number, $d_5$, LSI, Voronoi density and asphericity, pair-entropy fingerprint and its local average.*
*SolOrder: density, coordination_number, smooth_coordination, d5, local_structure_index, voronoi_density, voronoi_asphericity, eFingerprint.*

## 3.3 Angular and tetrahedral parameters

*Plan: Orientational and translational tetrahedral order, water–water angular distribution, bond-angle, angle-histogram and angular Fourier parameters, local-neighborhood A1–A3.*

*SolOrder: tetrahedral_order, translational_order, water_angular_distribution, bond_angle_parameter, angle_histogram_parameter, angular_fourier, neighborhood_parameters.*

## 3.4 Hydrogen-bond and topology-based parameters

*Plan: H-bond counts, $F_4$ (including the hydrogen-selection pitfall found during validation), Russo–Tanaka $\zeta$, smallest-ring statistics of the H-bond network (note that periodic wrap-around paths must be rejected in small cells), graph descriptors.*
*SolOrder: hydrogen_bonds, F4, zeta, ring_statistics, graph_descriptor.*

## 3.5 Bond-orientational order

*Plan: Steinhardt $q_l$, $w_l$, normalized $\hat{w}_l$, averaged variants, crystalline connections, CHILL and CHILL+.*
*SolOrder: steinhardt, crystalline_connections, chill, chill_plus.*

## 3.6 Template and symmetry-based parameters

*Plan: Polyhedral template matching, centrosymmetry, neighbor-distance parameter $D$.*
*SolOrder: template_matching, centrosymmetry, neighbor_distance_parameter.*

## 3.7 Molecular and electrostatic parameters

*Plan: Quadrupole-related $Q_T$; when orientational information beyond oxygen positions is necessary.*
*SolOrder: quadrupole_parameter.*

## 3.8 Summary table

Table 3.1 lists every parameter in the benchmark with the settings used there (benchmarks/registry.py). "Smooth" means continuous in the atomic coordinates, the property required of a collective variable.

Table 3.1: Local order parameters in the benchmark, with the settings used there. All O–O first-shell cutoffs are 3.5 Å. [TODO: add cost column after the scaling runs.] ?: reference to add.

| Parameter | Group | Neighbors | H | Smooth | Ref. |
|---|---|---|---|---|---|
| CN | radial | O within 3.5 Å | | | – |
| Smooth CN | radial | switching function, $r_0 = 3.5$ Å | | ✓ | – |
| $d_5$ | radial | fifth-nearest O | | | [1] |
| LSI | radial | O within 3.7 Å and next | | | [2] |
| Voronoi density, asphericity | radial | periodic Voronoi cell | | | – |
| Entropy $s$, $\bar{s}$ | radial | $g(r)$ to 5 Å; average to 5 Å | | ✓ | [3] |
| $Q$, $\bar{Q}$ | angular | 4 nearest O; average 3.5 Å | | | [4] |
| $S_k$ | angular | 4 nearest O | | | [5] |
| Bond angle $B$ | angular | O within 3.5 Å | | | ? |
| Angular Fourier ($a = 3$) | angular | O within 3.5 Å | | | ? |
| Angle histogram $H_1$–$H_3$ | angular | O within 3.5 Å | | | ? |
| $A_1$–$A_3$ | angular | O within 3.5 Å | | | ? |
| $F_4$ | H-bond | O within 3.5 Å, outer H | ✓ | | [6] |
| $N_{\mathrm{HB}}$ | H-bond | 3.5 Å, 30° | ✓ | | [7] |
| $\zeta$ | H-bond | H-bond partners; O to 5 Å | ✓ | | ? |
| Ring fractions (4–8) | H-bond | H-bond network, rings $\leq 10$ | ✓ | | ? |
| $q_3$, $q_4$, $q_6$ | bond-orient. | O within 3.5 Å | | | ? |
| $\bar{q}_3$, $\bar{q}_4$, $\bar{q}_6$ | bond-orient. | averaged over 3.5 Å | | | [8] |
| $\hat{w}_4$, $\hat{w}_6$ | bond-orient. | O within 3.5 Å | | | ? |
| Crystalline connections | bond-orient. | $q_6$ dot product $> 0.5$ | | | [9] |
| CSP | template | 12 nearest O | | | [10] |
| Neighbor distance $D$ | template | O within 3.5 Å | | | – |
| CHILL / CHILL+ | labeler | 4 nearest / 3.5 Å, $q_3$ | | | [11, 12] |
| PTM | labeler | diamond templates | | | [13] |

# Chapter 4

# Descriptors and machine-learned approaches

*Plan: Critical, not promotional: what each method needs (training data, labels, model), what it reports, and how it was validated.*

## 4.1 High-dimensional descriptors

*Plan: Descriptor requirements (invariance, uniqueness, continuity, conciseness, cost); SOAP, ACSF, MBTR/LMBTR, Coulomb/sine/Ewald matrices.*
*Reuse: background/background.tex sections "Descriptor" and "SOAP".*
*SolOrder: soap, atomic_descriptor.*

## 4.2 Supervised classifiers

*Plan: DeepIce, GCIceNet, PointNet-style and equivariant networks. Report accuracies with the dataset they refer to.*
*Reuse: background/background.tex sections "DeepIce" and "GCIceNet".*

## 4.3 Unsupervised and dimensionality-reduction approaches

*Plan: PCA, clustering, autoencoders on SOAP; mention the author's SOAP autoencoder only as a pointer to separate work.*
*Reuse: conclude/conclude.tex, condensed to one paragraph and one figure at most.*

## 4.4 Data-driven collective variables

*Plan: Deep-LDA, Deep-TICA, OPES/metadynamics with learned CVs for ice nucleation.*

[TODO: literature.]

# Chapter 5

# A unified benchmark

All parameters in Table 3.1 were computed with one implementation (SolOrder), one set of settings fixed before any comparison, and one analysis pipeline. Differences between parameters therefore reflect the parameters themselves rather than codes, cutoffs or water models. The design, code and results are in the benchmarks/ directory of the SolOrder repository. [TODO: Results below are from the ideal crystals and the 220 K seed test; update when the thermalized high-pressure phases (Sec. 5.1) are available.]

## 5.1 Datasets

**Ideal crystals.** Eighteen phases were generated with GenIce2 [TODO: cite]: Ih, Ic, II, III, IV, V, VI, VII, VIII, IX, XI, XII, XIII, XIV and XVII, and the empty sI, sII and sH clathrate frameworks. Each proton-disordered phase was generated from five random seeds, giving 89,100 molecular environments in total. Ice XVI was excluded because GenIce gives it the same oxygen positions as the empty sII framework. Every atom was displaced by Gaussian noise of 0.02 Å, about twice the coordinate precision of the files, so that two phases count as separable only at a stated resolution rather than through rounding differences. Five pairs relate proton-ordered and proton-disordered forms of one oxygen framework: Ih/XI, III/IX, V/XIII, VII/VIII and XII/XIV.

**Thermal and quenched TIP4P/Ice.** Bulk Ih, Ic and stacking-disordered ice, and spherical Ih, Ic and stacking-disordered seeds of 50, 100 and 200 molecules embedded in liquid, were simulated at 220 K (TIP4P/Ice [TODO: cite], NVT). Each of the twelve systems contributes ten thermal frames and the same frames after energy minimization, 834,040 molecular environments in all. Every molecule carries a ground-truth label from the construction: seed membership, interior position, and

hexagonal or cubic stacking of its layer. The liquid reference consists of molecules more than 8 Å beyond the seed radius.

[TODO: (i) Thermalized trajectories of each phase and of liquid at matched $T, P$; (ii) amorphous ices; (iii) a second water model.]

*SolOrder: benchmarks/compute.py, scripts/generate_genice_fixtures.py.*

## 5.2 Separability metric

For a scalar parameter and two populations $A$ and $B$ we use the two-sample Kolmogorov–Smirnov statistic

$$D_{AB} = \sup_x |F_A(x) - F_B(x)|\,, \tag{5.1}$$

where $F_A$ and $F_B$ are the empirical cumulative distributions of the parameter over the molecules of each population. $D$ needs no binning, which matters for the near-delta distributions of ideal crystals. $(1+D)/2$ is exactly the balanced accuracy of the best single threshold, so it is reported as an accuracy between 0.5 (no separation) and 1 (perfect separation). Molecules within one frame are correlated, so 95% confidence intervals come from a bootstrap over frames rather than over molecules. Labelers (CHILL, CHILL+, PTM) are scored instead against the ground-truth labels.

## 5.3 Phase-by-phase separability

**Ideal crystals.** Of the 153 phase pairs, 150 are separated at a balanced accuracy of at least 0.95 by at least one parameter (Fig. 5.1). Three are not: Ih/XI (best 0.87, smooth CN), V/XIII (best 0.83, $\bar{q}_4$) and sI/sII (best 0.92). The first two are proton order/disorder partners whose oxygen frameworks barely differ. No parameter that uses hydrogen positions ($F_4$, $\zeta$, $N_{\mathrm{HB}}$, ring fractions) separates them either. The other partner pairs are separated only because the ordered cells are slightly distorted: III/IX by $\bar{s}$ alone, VII/VIII by 17 parameters and XII/XIV by five. [TODO: Check whether the GenIce cell distortions match experimental lattice parameters before interpreting these.] The most broadly useful parameters are $\bar{s}$ (93% of pairs at $\geq 0.95$), Voronoi asphericity (90%), Voronoi density and $\bar{Q}$ (89%). $N_{\mathrm{HB}}$ separates no pair, because every molecule in every ideal crystal has four hydrogen bonds; $A_3$ separates none either.

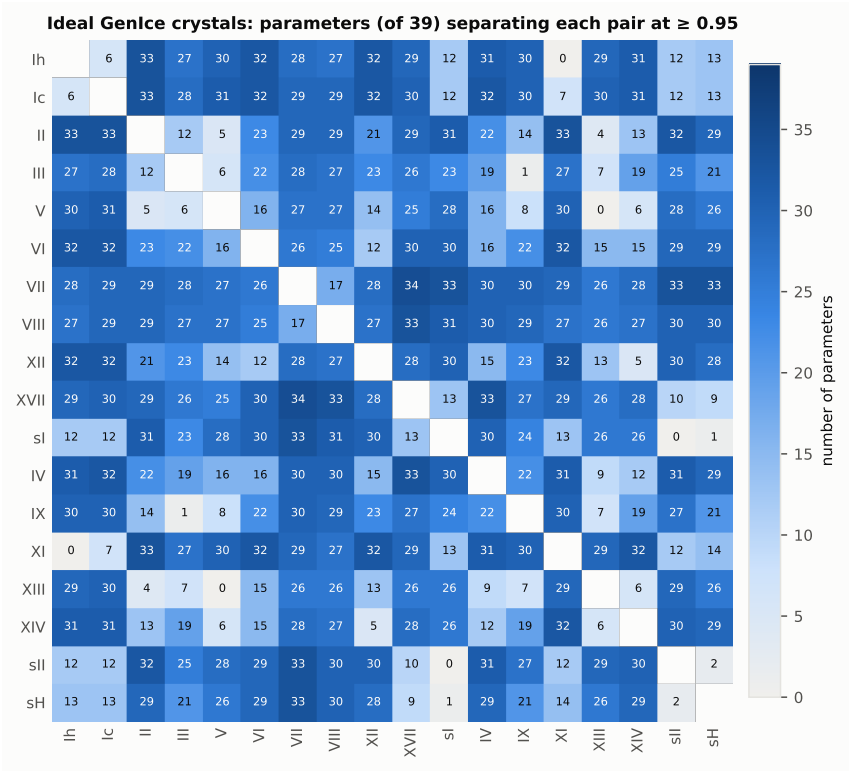



Figure 5.1: Number of parameters (of 39) that separate each pair of ideal crystals with a balanced accuracy of at least 0.95. The full parameter × pair matrix is in the Supplementary Information (S1_separability_ideal_all_pairs).

**Thermal and quenched structures.** At 220 K the picture changes (Fig. 5.2). For ice versus liquid, the best parameters are locally averaged: $\bar{s}$ (1.00), $\bar{q}_6$ (0.99) and $\bar{Q}$ (0.98), compared with 0.85, 0.78 and 0.84 for their unaveraged forms. The 6-ring fraction (0.99) is the best unaveraged parameter. The cutoff coordination number (0.62) and $N_{\mathrm{HB}}$ (0.53) are nearly useless. Energy minimization restores most of the separation removed by thermal motion: $d_5$, LSI and $\zeta$ rise from 0.83–0.84 to 1.00.

Ih and Ic are separated at 220 K only by $\bar{q}_3$ (1.00), $\bar{q}_4$ (1.00), $F_4$ (0.94), crystalline connections (0.93), $\bar{q}_6$ (0.78) and CSP (0.72). Every other parameter is at 0.61 or below. Ring statistics cannot distinguish them, since both contain only 6-rings. $\bar{s}$ separates Ih from Ic perfectly in ideal crystals but only at 0.61 at 220 K.

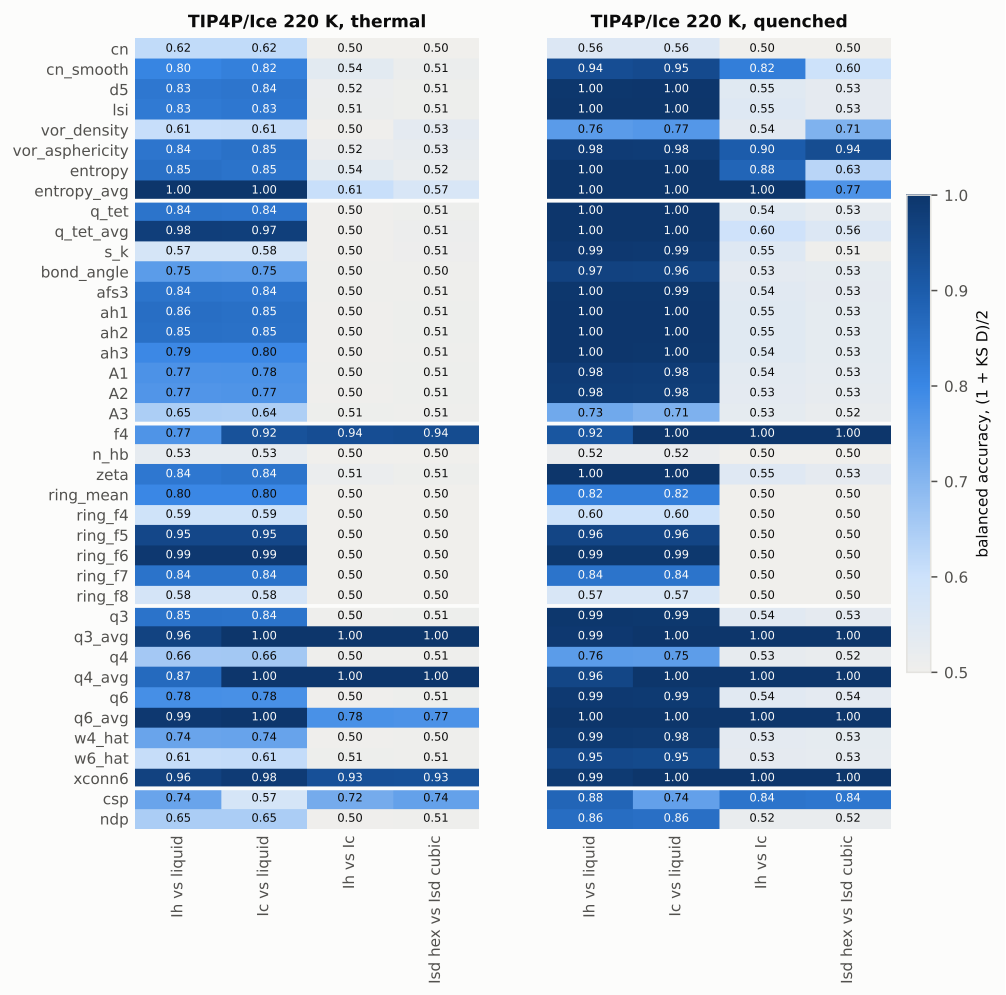



Figure 5.2: Best single-threshold balanced accuracy for ice versus liquid, Ih versus Ic, and hexagonal versus cubic layers in stacking-disordered ice, TIP4P/Ice at 220 K. Left: thermal frames; right: the same frames after energy minimization.

**Redundancy.** Many parameters carry nearly the same information. Grouping parameters whose Spearman rank correlation exceeds 0.8 in absolute value reduces the 39 columns to 22 groups (Fig. 5.3). The largest group joins $Q$, the bond angle, the angular Fourier term, the three angle-histogram amplitudes, $q_3$, $q_6$ and $\hat{w}_4$. Other groups join $\bar{q}_3$, $\bar{q}_4$, $\bar{q}_6$ and crystalline connections; $d_5$, LSI, $s$ and $\zeta$; $\bar{Q}$ and $\bar{s}$; and the 5- and 6-ring fractions.

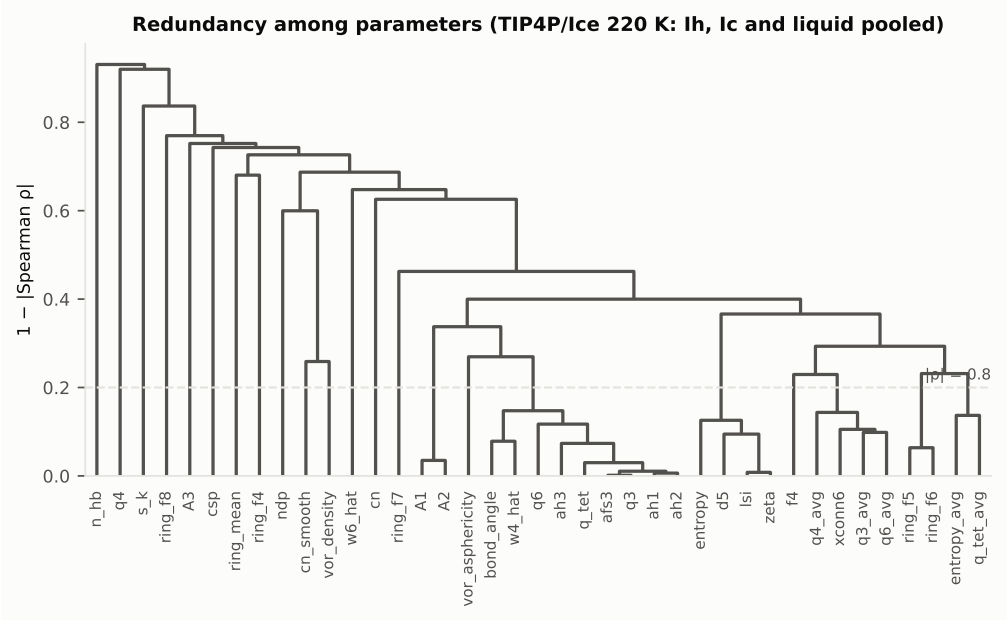



Figure 5.3: Hierarchical clustering of parameters by $1 - |\rho|$, with $\rho$ the Spearman correlation over thermal Ih, Ic and liquid molecules at 220 K. The dashed line marks $|\rho| = 0.8$.

## 5.4 Ice labelers against ground truth

CHILL+ labels 99.7% of thermal bulk Ih correctly, while the original CHILL labels only 70.7%, plausibly because its narrow eclipsed-bond window misses thermally distorted hexagonal environments, the motivation for the wider window in CHILL+ [12]. [TODO: confirm from the eclipsed-bond counts.] PTM with the default RMSD cutoff labels 97% of bulk ice correctly and never labels liquid as ice, but it misses most molecules in small seeds (Fig. 5.4). In the seeds, all three labelers recover only part of the interior: CHILL+ finds 11% of interior molecules of a 50-molecule Ih seed with the correct polytype, and 58% for a 200-molecule seed. CHILL+ is best for Ih and stacking-disordered seeds; for Ic seeds CHILL is level or slightly ahead. False-positive rates in the far liquid are at most 1% for all three.

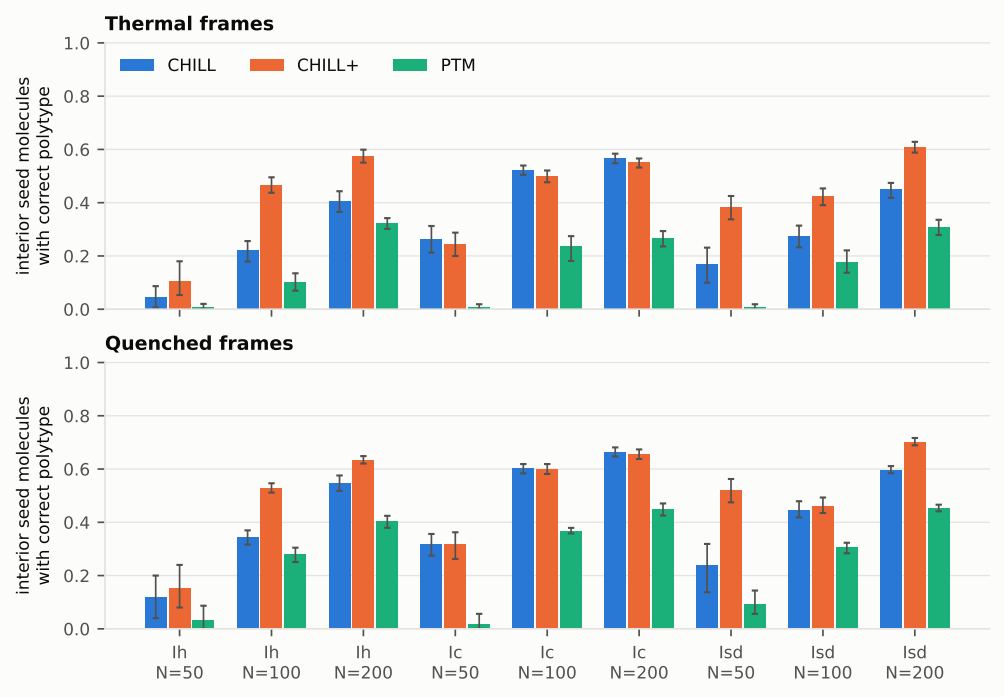



Figure 5.4: Fraction of interior seed molecules labeled with the correct polytype by CHILL, CHILL+ and PTM, for Ih, Ic and stacking-disordered (Isd) seeds of $N$ molecules in TIP4P/Ice liquid at 220 K. Error bars: 95% bootstrap intervals over frames.

[TODO: Caveats to state: frames are 1 ps apart, so the bootstrap intervals are likely too narrow; the liquid reference is supercooled; ideal-crystal results depend on GenIce cell geometries.]

## 5.5 Sensitivity to neighbor definition and water model

*Plan: Cutoff scans; N-nearest vs. cutoff; TIP4P/Ice vs. mW where available.*

## 5.6 Computational cost

*Plan: Time per molecule vs. system size, serial and parallel.*
*SolOrder: n_jobs and backend options (parallel.py).*

## 5.7 Application: following an interface or nucleation event

*Plan: Per-molecule labels along one trajectory from CHILL+, $q_6$-based, PTM and a descriptor-based method; agreement and disagreement.*

# Chapter 6

# Practical guide

*Plan: Decision table: question (ice vs. liquid, Ih vs. Ic, hydrate vs. ice, high-pressure polymorphs, collective variable for sampling) → recommended parameter(s), settings and caveats.*

Table 6.1: Provisional recommendations from the benchmark so far (ideal crystals and TIP4P/Ice at 220 K). [TODO: revise with the thermalized high-pressure phases.]

| Question | Best so far | Avoid |
|---|---|---|
| Ice vs. liquid | $\bar{s}$, $\bar{q}_6$, $\bar{Q}$; 6-ring fraction if no averaging is wanted | CN, $N_{\mathrm{HB}}$ |
| Ih vs. Ic | $\bar{q}_3$, $\bar{q}_4$; $F_4$ if hydrogens are available | ring statistics, unaveraged parameters |
| Per-molecule labels in seeds | CHILL+ (Ih, stacking-disordered), CHILL or CHILL+ (Ic) | CHILL for Ih; PTM for seeds under ∼100 molecules |
| Distinguishing crystal phases | $\bar{s}$, Voronoi asphericity and density, $\bar{Q}$ | $N_{\mathrm{HB}}$, $A_3$ |
| Proton-ordered vs. disordered | no local parameter tested is reliable | – |

## 6.1 Software

*Plan: SolOrder alongside freud, OVITO, PLUMED, pyscal and MDAnalysis: coverage, parallelism, differentiability, licensing.*

## 6.2 Reproducibility checklist

*Plan: What to report when using an LOP: neighbor rule, cutoff, averaging, water model, thresholds and their calibration.*

# Chapter 7

# Outlook

*Plan: Open problems: proton-ordered partners, amorphous ices, interfaces and confinement, transferability of learned models across water models, LOPs as differentiable CVs.*

## Data and code availability

*Plan: SolOrder version, benchmark scripts and datasets (DOI).*

## Acknowledgments

*Plan: Funding, computing resources.*

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